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标题	Decarboxylative Cross-Coupling Enabled by Fe and Ni Metallaphotoredox Catalysis
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出处	Journal of the American Chemical Society 2024
DOI	10.1021/jacs.4c09621
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC11528444/
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摘要（原文节选）

Decarboxylative cross-coupling of carboxylic acids and aryl halides has become a key transformation in organic synthesis to form C(sp2) – C(sp3) bonds. In this report, a base metal pairing between Fe and Ni has been developed with complementary reactivity to the well-established Ir and Ni metallaphotoredox reactions. Utilizing an inexpensive FeCl3 cocatalyst along with a pyridine carboxamidine Ni catalyst, a range of aryl iodides can be preferentially coupled to carboxylic acids over boronic acid esters, triflates, c ⋯

数据表 · Table 1（12 行，含数值行 12 行）

Selected Optimization and Control Data for the Fe and Ni Decarboxylative Couplinga

entry	deviation from above	yield 1 (%)
1	none	86
2	no Fe catalyst	19
3	no Ni catalyst	0
4	no ligand	0
5	no light (60 ° C)	0
6	no TBAI	73
7	with 0.5 equiv of phthalimide	90
8	Cs2CO3 instead of iPr2NEt	9
9	KI instead of TBAI	72
10	TBACl instead of TBAI	31
11	427 nm instead of 390 nm	62
12	MeCN instead of 1,4-dioxane	67

脚注（逐字）：aReactions were performed on a 0.5 mmol scale. Acid (1.3 equiv), aryl halide (1 equiv), iPr2NEt (2 equiv), TBAI (1 equiv), FeCl3 (3 mol %), Ni(NO3)2 · 6H2O (3 mol %) and 4-tBuPyCamCN (3 mol %) were added to 10 mL of 1,4-dioxane, and the mixture was sparged with N2 for 5 min. The mixture was irradiated

正文对应句（逐字）：Organohalide and carboxylic acid scope for the Fe and Ni decarboxylative arylation reaction. Yields shown are isolated yields after purification. Reactions were performed on a 0.5 mmol scale with acid (1.3 equiv), aryl halide (1 equiv), iPr2NEt (2 equiv), TBAI (1 equiv), FeCl3 (3 mol %), Ni(NO3)2 · H2

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